

# A Survey on AI-Based Chatbots in Customer Service

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**Abstract**—Companies everywhere are trying to keep up with the demand for fast, cheap customer service. Human support teams often get overwhelmed by the sheer volume of questions, leading to long hold times and frustrated users. Because of this, many businesses are turning to AI chatbots. Thanks to recent updates in machine learning, natural language processing, and large language models (LLMs), these bots are now much better at figuring out what users want and giving answers that actually make sense.

This paper provides a broad survey of how AI chatbots are used in customer service today. We look at how the technology has evolved, how different systems are built, and how they are classified. We break down the main architectures—from simple rule-based and retrieval bots to advanced generative and LLM-powered setups. The study checks how well these models actually perform in industries like banking, e-commerce, and healthcare. We also review the standard metrics used to test them, such as response speed, accuracy, and user satisfaction.

Even with all the recent progress, chatbots still have real problems. They often lose track of long conversations, struggle to personalize their answers, and raise serious concerns about data privacy. We discuss these exact roadblocks and point out where the industry is heading next, like pairing AI with human agents. The main goal of this survey is to give a clear picture of current chatbot technology and highlight exactly what needs fixing to make these systems more reliable and user-friendly.

**Keywords**—Artificial Intelligence, Chatbots, Customer Service Automation, Natural Language Processing, Large Language Models

## I. EVOLUTION OF CHATBOTS

There has been a significant improvement in the quality of chatbots, from simple rule-based chatbots to advanced chatbots that are aware of user context. As artificial intelligence has progressed, chatbots have also significantly improved over time, utilizing AI, machine learning, deep learning, and natural language processing (NLP).

Turing Test started the theoretical work for conversational chatbots, which tried to answer whether a machine can talk like human [12].

Joseph Weizenbaum developed ELIZA, the first chatbot, in the 1960s. It was a rule-based chatbot that tried to simulate a conversation with a psychotherapist, more precisely by mimicking the comprehension of the interlocutor, although it was unable to completely understand what was being said [1]. It was shown by the program led by Joseph Weizenbaum that even without actual semantic understanding of the input, a similar feeling of empathy can be developed with simpler methods like pattern recognition and pronoun substitution [13]. The main issue with this chatbot was that there were limited options available. Most chatbots then were ineffective because they relied solely on keyword recognition [4].

In the 1970s, systems like PARRY went one step further with this idea by trying to imitate more specific human behaviors, like those of a paranoid patient. Though it was better and more advanced than previous solutions, PARRY still required a fixed set of rules with negligible understanding of language [1]. These technologies showed the possibilities of chatbot technology but also showed that more flexible and intelligent systems are required.

The era of the 1980s and 1990s proved to be another major step in the advancement of chatbot technology. The development of artificial intelligence and machine learning techniques provided an opportunity for more refined chatbot technology. During this period, a shift was made in chatbot technology from utilizing simple methods like decision trees, basic neural networks, and hidden Markov models to using more refined methods like natural language processing, which helped make chatbots aware of the user context of the input [4]. Another major advancement during this time was that natural language processing technology improved significantly, which made chatbots more aware of the basic intent behind an input and gave them a better grasp of parts of speech.

During the 2000s, more practical and widely used chatbots were developed, like SmarterChild, which was used on messaging platforms and provided general additional information

like news and weather updates, which were real-time events [1]. It was during this era that chatbots became more visible commercially, and features like understanding of user tone and preference were developed [2].

A major leap in chatbot technology was observed in the 2010s with the advancement of deep learning and advanced neural networks. Technologies like recurrent neural networks (RNNs) and convolutional neural networks (CNNs) were developed, which enhanced the capabilities of chatbots by enabling them to identify patterns in data. The shift towards deep learning enabled algorithms to process massive datasets and shifting conversational agents away from rigid and hard-coded rules [14]. During this time, virtual assistants like Siri were developed using natural language processing and machine learning techniques and proved to be an initial effort toward more user-specific chatbots [1]. The addition of advanced speed recognition and natural language processing set a new standard for interactive customer service via voice-activated system [15]. For processing complex queries with good accuracy, large-scale AI systems like IBM Watson were also developed [2]. IBM Watson demonstrated the working ability of AI to process natural language queries with high accuracy which highlighted its usefulness for problems in specialized domains [16].

In the most recent phase of chatbot technology advancement, transformer models and large language models (LLMs) are being used directly in the development of chatbots. This structure allowed AI to use attention mechanisms which can be used in both directions, which allowed systems to understand the complete context of an input rather than processing words one by one in a sequence [17]. More human-like responses can be produced in a conversation with a chatbot using the latest GPT models. The creation of LLMs such as GPT-3 has set the benchmark for conversational artificial intelligence because these models are trained on huge amounts of data to create natural language that is similar to human interaction [18]. Now chatbots also understand user intent and adapt to various situations, along with understanding the context. Advanced features like sentiment analysis, personalization, and learning have been added to chatbots, which is helping improve user experience [4].

With frameworks like LangChain, LLMs can be used with external sources of information, thereby helping chatbots produce personalized, real-time, as well as domain-specific responses. With the latest models, there is a shift from knowledge-based to knowledge-driven systems that can be used to develop complex chatbots like customer service automation [3].

In summary, the evolution of chatbots can be observed in four major phases: rule-based chatbots, AI-based retrieval chatbots, ML-based chatbots, and the latest LLM-based chatbots. With every new phase, the shortcomings of the previous phase have been addressed to make chatbots more intelligent and usable. Today, chatbots are more aware of user context and offer a personalized experience.

## II. CLASSIFICATION

In present day customer support systems, chatbots are an important part because they allow automatic, scalable, and real-time communication between companies and consumers. The research that is currently being published indicates that there are several ways to approach chatbot classification, including application domain, algorithmic design, and functionality [11]. In customer service, the two most crucial classification dimensions are the service function and the underlying reasoning mechanism.

### A. Classification Based on Service Function

Chatbots are basically task-oriented systems that help customers with tasks like getting information, solving problems, or completing transactions.

1) *Chatbots for Getting Information (FAQ)*: These chatbots specialize in answering questions by getting relevant answers from an organized database. They are often used in customer service to answer common questions and give customers an overview of the products and services. Information retrieval-based frameworks make it easy to search through large stores of knowledge and have discussions based on search results [8].

- **Benefits:** Accurate for a limited set of questions; responds very fast; great for extensive use.

- **Limitations:** Not flexible enough when dealing with questions that aren't clear or are hard to answer; answers lack in-depth analysis and personal touch.

2) *Chatbots Focused on Tasks and Transactions*: Task-oriented chatbots are designed to do specific things, like making service reservations, taking payments, or changing user data. These systems mainly use entity extraction and intent detection techniques to map user input to pre-defined service workflows [7].

- **Benefits:** Automating routine customer service tasks well; connecting to enterprise backend systems.

- **Limitations:** Work processes are rigid; hard to handle tasks that don't go according to plan.

3) *Chatbots for Helping with Conversations*: Conversational chatbots can have multi-turn conversations that take into account the context and go beyond set tasks. These systems use machine learning and deep learning models to make conversations more like those between people, which keeps users interested. Generative conversational models can make responses that change based on what the user says [5].

- **Benefits:** Greater user interaction and a better experience for users; greater capacity to respond to complex questions.

- **Limitations:** The possibility that chatbots will give incorrect, or unhelpful responses.

4) *Hybrid AI-Human Chatbots*: Real world business applications for hybrid (AI-human) chatbot systems include combining an automated chatbot interface with a live human representative's interface. This allows the chatbot portion of the hybrid system to take care of answering commonly asked

questions in which the solution is well-known, so that the human representatives can focus on handling new and more complicated problems [21].

- **Benefits:** Higher-quality services; lower costs than using human representatives as long as there are adequate levels of supervision.
- **Limitations:** More complicated systems; dependency on smooth transitions from one part of the system to another.

### B. Classification Based on Intelligence Mechanism

Additionally to functional classifications, chatbots may also be classified according to how they process and produce their response.

1) *Rule-Based Chatbots:* Rule-based chatbots utilize rules and scripts. Most of these early versions of chatbots were implemented as part of initial Customer Support. They did however fail to work properly when users failed to follow the specific rules defined by the system [7]. Although rule-based systems present the advantages of clarity and organization through a structured decision-making procedure, they can be overly restrictive; thus ensuring that the established rules are followed is an essential element when operating within high-reliability environments [19].

2) *Retrieval-Based Chatbots:* With regard to Retrieval-Based Chatbots, the generated response is selected from a predefined list of possible answers. This style of chatbot provides both grammatically accurate answers and reliable results. However, they lack the ability to create fresh and new responses [5]. To enhance understanding of recursive conversation scenarios, advanced retrieval-based chatbots now apply framework based on sequential matching, which link prior responses together to determine the most suitable stored answer [20].

3) *Generative Chatbots:* Generative chatbots operate on Deep Learning methods (i.e. Encoder-Decoders). As such, they possess greater versatility and interaction capacity than retrieval-based chatbots. Additionally, generative chatbots have demonstrated superior performance rates compared to retrieval-based chatbots [5].

4) *LLM-Based Chatbots:* There have recently been significant advancements in field of Large Language Models. These advances have provided for enhanced functionality of chatbots. Specifically, LLM-based chatbots are capable of contextualizing information and providing natural and accurate conversations. Furthermore, LLM-based chatbots may function across multiple domains and serve as the current best practice in technology [6]. The advancement of LLM-based bots has created a need for frameworks (such as Retrieval-Augmented Generation (RAG)) designed to address limitations in this field such as Hallucinations [22].

The classification has also indicated that in customer service, the technology is moving from simple question answering to complete automated service and personal conversation. However, both categories have their own trade-off's when it comes to computational complexity, flexibility, and reliability.

## III. ARCHITECTURE

### A. Architecture of Rule-Based Chatbots

The first conversational systems were rule based chatbots. These type of chatbots use fixed, predictable processes that are primarily based on a developers defined logic. To break it down into simpler terms, rule based chatbot architectures can be broken down into two primary areas: input and response generation. Both of these steps are heavily governed by manually written rules [7].

The input area will take users input (what they typed) and through the use of some basic pattern matching techniques determine how to interpret the users intent. Typically developers will create these rules using markup language such as AIML(Artificial Intelligence Markup Language). The developer will define an intent within the AIML file. When a user inputs something the system will scan the text for the intent(s) which the developer has defined and then map those intent(s) to one or multiple decision trees/rules. Think of this process similar to "if-then" statements. When the system identifies a match with one of its rules it will pull a pre-defined response template from a database. Since this type of architecture is highly predictable and controlled, rule-based architectures work extremely well for tasks which are structured such as answering FAQ's, customer support inquiries etc. Since rule based architectures don't require large amounts of data, they are able to rely on the information given by subject matter experts and perform at relatively low computational costs [7,23,24].

Beginning with breaking the architecture, there will typically be the following modules:

- **Input Processing:** The input processing module performs the basic tasks such as tokenization and simple syntax parsing.
- **Rule Engine / Pattern Matcher:** The pattern matcher/rule engine is the "brain" of the operation; it identifies how well the user's request matches a pre-defined set of scripted rules.
- **Dialogue Manager:** The dialogue manager manages the flow of a conversation by making decisions about what should happen next at each point in a conversation through finite state or decision trees.
- **Response Repository:** A simple database storing all the canned responses.

These architectures suffer from being extremely inflexible. The difficulty in scaling these systems is that developers must identify every way a user could ask a question before implementing the solution. Add to this the presence ambiguity in wording, or the unexpected question, and the chatbot quickly deviates from its intended course. When it cannot find a matching rule it falls back to a default response which quickly becomes frustrating for users when used in real world applications [7]. Furthermore, because they contain no form of learning capabilities, they cannot evolve over time based upon user feedback. Their functionality is confined to very narrow

domains – though, in doing so, they provide reliability and ease of auditing [9].

### *B. Architecture of Retrieval-Based Chatbots*

Retrieval-based chatbots provided a data driven methodology to replace rigid rules. Unlike forced scripting of every potential interaction, retrieval-based chatbots utilize both machine learning and information retrieval (IR) to select the optimal response(s) from a predefined database.

The overall architecture for these retrieval-based chatbots includes a three-layered structure including natural language understanding (NLU), a retrieval layer, and a response selection layer. In general terms, the NLU transforms the user's text into a structurally organized representation (such as feature vector or neural embedding). Next, the IR layer searches the knowledge base to generate a collection of potential responses based upon similarities calculated via various methods. After the candidate list has been generated the candidate(s) will be scored by either a machine learning model or a number of heuristics, and therefore selected as the "winner" of the system.

Generally, a standard retrieval-based architecture will include the following elements:

- **Pre-processing and Vectorization of the Raw Text:** Converts raw text to vectors for use as input to models through various representations such as TF-IDF or neural embeddings.
- **Query Understanding Module:** Extracts the user's intent and contextual cues from the input query.
- **Information Retrieval Engine:** Searches the knowledge base or large indexed dataset to retrieve all candidate response strings that match the search criteria. The current computing environment has utilized vector databases (such as FAISS) to store semantic embeddings which support fast retrieval of candidate answers as well as reduced memory usage [27].
- **Ranking and Selection Module:** Evaluates each candidate response string and identifies the best candidate response string.
- **Dialogue Context Handler:** Tracks dialogue state during a multi-turn conversation.

This class of bots utilize IR techniques to enable rapid searching of massive datasets. Examples include how public service chatbots combine IR techniques with NLP techniques to help users navigate complex, multi-turn information retrieval processes [8]. They provide a much better user experience because they allow both keyword searches and natural phrase searches compared to earlier rule-based systems.

Deep learning provided another significant improvement to this architecture. By employing neural models including CNNs, LSTMs, and RNNs to identify relationships between questions and answer candidates, the accuracy of the matching process was significantly increased [5]. However, the same fundamental limitation applies; they cannot create original content.

From an architectural perspective, there is a significant trade-off. On one hand, you gain complete control over your precision and grammar since you have complete control over the candidate responses (i.e., since responses were written by humans). On the other hand, you lose total creative freedom and flexibility. In addition, performance is strictly dependent on the quality of your dataset. If the correct answer does not exist within the dataset, then the bot will be unable to respond appropriately. As well as being time-consuming for expansion into another area (i.e., large annotated data sets), curation and maintenance of those data sets is necessary for expanding into another area.

### *C. Architecture of Generative and LLM-Based Chatbots*

In addition to the above mentioned paradigms; a major paradigm shift occurred when generative architectures emerged. These systems do not retrieve a response from a database that includes all possible answers. Rather, these systems create synthetic text using neural networks. This creates entirely unique responses for users. Since the system can generate an infinite number of unique responses based upon user input; these systems have been successful at handling a wide variety of conversations, including very lengthy ones.

1) *Encoder-Decoder and Neural Generative Architectures:* The first generation of generative AI models were built around sequence-to-sequence (Seq2Seq). For example, an encoder compressed the input into a hidden or latent representation. A decoder then decompressed each token in the latent space to create a response. Although this provided a way for chatbots to provide novel responses, and improve their ability to respond to many different types of conversations; there were still some limitations. In particular, early generative models suffered from poor contextual retention over longer conversations. Therefore, generative models were limited to producing coherent responses. Additionally, the need for large amounts of training data made them difficult to train.

2) *Transformer and LLM-Based Architectures:* Next was the development of the transformer model which changed the architecture of chatbots once again. Self-attention mechanism in large language models (LLM) allow for tracking long range dependencies in conversation [6]. Therefore, because LLM's are trained on such large text corpuses; they generalize extremely well across many different areas of study. Furthermore, since LLM's are trained on so much text; they produce responses that resemble those written by humans.

The typical LLM stack looks something like this:

- **Input Processing and Tokenization:** Breaks the user's text down into token sequences.
- **Embedding Layer:** Maps those tokens into a high-dimensional vector space.
- **Transformer Blocks:** Does the heavy lifting, using self-attention to capture the nuanced contextual relationships across the text.
- **Decoding Module:** Predicts and generates the output tokens autoregressively.

- **Post-processing and Safety Filters:** Gives the final text a safety check and enforces formatting constraints.

These models are exceptionally good at keeping track of multi-turn dialogues and understanding subtle language nuances. Unlike the older architectures, LLMs fuse language understanding and generation into one unified framework, completely ditching the need for separate rule or retrieval engines.

3) *Hybrid and Enterprise Architectures:* In real-world enterprise environments, however, you rarely see an LLM running entirely on its own. Hybrid setups are the standard today.

Take Retrieval-Augmented Generation (RAG), for example. It pairs the creative power of an LLM with external knowledge bases to ensure factual accuracy and domain relevance. Enterprise-grade systems layer on additional modules for security, scaling, and backend API integration to support robust, real-time customer service [10]. Furthermore, attaining high availability in such enterprise implementations depends significantly on incorporating unique fine-tuning approaches along with advanced context management methods into the hybrid framework itself [28]. Frameworks like LangChain have become incredibly popular here, acting as the orchestration glue that connects LLMs to live business data, external embeddings, and specialized tools [3],[26].

The leap in capability here is undeniable. Generative and LLM architectures offer:

- The ability to craft fresh, contextually perfect responses on the fly.
- A massive leap in conversational fluency and personalization.
- Easy scalability across different domains without the nightmare of writing new rules.

But the trade-offs are just as steep. These architectures are computationally exhausting and highly dependent on massive amounts of clean data. They also constantly risk “hallucinating” incorrect facts or mirroring biases found in their training data. Navigating these reliability and ethical issues remains the biggest hurdle for deploying them safely in sensitive industries [9].

#### IV. APPLICATIONS IN CUSTOMER SERVICE

The introduction of Artificial Intelligence (AI) into Customer Service is no longer an experimental cost saving initiative; it has become a fundamental part of Enterprise Digital Transformation Strategies [31]. The advancement from scripted, rule based chatbots to autonomous ‘agentic’ systems permits businesses to address both routine inquiries as well as complex multiple step work flows which have historically required Human Intervention [32]. Literature suggests that the use of chatbots has predominantly been seen within Retailing, Digital Banking, Telecommunications & Healthcare [33].

##### A. Industry-Specific Applications

1) *Retail & Ecommerce:* The retail & ecommerce space has the largest number of implementations with conversational AI. In this area, AI chatbots can be used to create a seamless

experience between digital convenience and the personal interactions you would receive at a physical store [34]. Based upon customer purchasing history and usage patterns, chatbots can present personalized product suggestions, allow customers to track orders, and give them options for returns [35]. Through the implementation of AI chatbots, retailers can offer customer support while promoting additional purchases. Additionally, chatbots are being implemented across various social media platforms (such as Facebook Messenger & WhatsApp) to engage with consumers where they prefer to communicate [36]. In addition to increasing revenue through chatbot engagement, companies can utilize AI chatbots throughout the after-sales process to automate activities such as tracking orders, returning items and providing delivery status updates. The shift towards RAG-based systems will ensure that the responses provided via chatbots are generated using current inventory levels and shipping schedules reducing the possibility of incorrect or inaccurate information being communicated to the end consumer [37]

2) *Banking and Finance:* BFSI is the second largest application of AI Chatbots; BFSI has some of the highest-risk applications of chatbots with an emphasis on accuracy, data-privacy and regulatory-compliance [38]. BFSI has taken the chatbot technology from being just an FAQ engine to a full-service financial advisor and transactional agent [39]. Banks are utilizing chatbots as virtual customer support agents. Examples include Erica, the Bank of America’s chatbot and Eno, which is Capital One’s chatbot. These chatbots utilize NLP to create conversationally-similar dialogues between humans and computers. Additionally, they alert users when a suspicious transaction occurs and also allow users to view their personal financial reports [40]. In addition to improving financial reporting and alerts to consumers regarding possible fraud, banking organizations are using chatbots to improve real time security. They achieve this goal by analyzing consumer’s past purchasing behaviors and identifying anomalies. Once identified, these anomalies can be used to send users instant messages that may prompt them to take action against potential fraudulent activity [41]. As well, there are examples where banks use chatbots to evaluate credit-worthiness. This includes evaluating a consumer’s historical purchase habits to establish whether or not the consumer should be eligible for a loan prior to the bank conducting a formal evaluation of the consumer’s credit worthiness [42].

3) *Telecommunications:* Telecommunications providers use chatbots and Interactive Voice Response Systems (IVRs) together to develop greater synergy within their customer service models. These chatbots are designed to employ NLP technologies to assist in processing voice-based inquiries from consumers and provide them with relevant answers to those inquiries based upon the type of inquiry [43]. Singh (2023), demonstrated how telecommunication providers could benefit from using both IVR and chatbots for customer service.

4) *Travel, Tourism, and Hospitality:* Within the service industry, chatbots provide a means of reducing the workload associated with servicing customers by automating tasks such

as booking schedules and modifying bookings. Anthropomorphism has been a major area of study within Travel-AI research. Anthropomorphism is defined as the attribute of giving a computer system characteristics of human beings including a name, a visual representation and an ability to speak in an empathic manner. Research studies indicate that people are more likely to book reservations with a company if they interacted with an AI powered booking system that utilized anthropomorphic design features than interacting with an AI-powered booking system without those features. Similarly, studies demonstrate that when individuals are provided with AI-powered booking systems that incorporate anthropomorphic design elements, individuals will report a higher degree of satisfaction when interacting with those systems compared to when they do not [44].

5) *Healthcare*: Scalable, continuous availability and cost reductions through 24/7 AI chatbot patient support for organizations with limited medical resources and increasing costs, though current clinical data on efficacy are still evolving [45]. Chatbots are being utilized widely in various areas including; Symptom Triage, Administrative Scheduling, Mental Health Support etc. [46] AI-Powered Symptom Checkers act as a primary entry-point for many healthcare professionals. Companion Bots and Smart Monitoring Systems utilize AI to monitor patients' condition in Real-Time and Predict potential Health Emergencies such as Diabetic Sugar Swings or Cardiac Instability before they occur [47]. LLMs exhibit particular aptitude to Simulate Therapeutic Techniques such as Cognitive Behavioral Therapy (CBT) and Cognitive Restructuring (CR) [48], Studies indicate that patients perceive them as Non-Judgmental 'Safe Spaces', Developing Trust and Relationship-Formation essential to Patient Loyalty [49]. Despite being able to Simulate Empathic Phrasing, and Socratic Questioning, there exists an 'Empathy Gap,' A significant Limitation to Effectiveness in High-Stakes Clinical Scenarios due to Limited Contextual Understanding [50].

#### *B. Operational Outcomes and Benefits*

Across Domains, it is apparent that AI chatbots aid in scaling-up Organizational Support Systems enabling Concurrent Interactions with no compromise to Quality of Service. As identified in studies some of the Key Operational Benefits of Chatbots include:

- **Continuous Availability**: The Chatbot is Continuously Available (24/7), regardless of Time Constraints. Always Ready to Help, and Available After Business Hours [51].
- **Cost Optimization**: Using chatbots for high-volume routine questions will result in a substantial reduction in Call Center Volumes and corresponding Human Resource Costs. Additionally, studies identify that Chatbots are capable of handling approximately 70% of Routine Questions [52].
- **Latency Reduction**: instantaneously obtaining information on FAQs and transactional items (such as password resets) is an important factor in building loyalty among consumers [43] [52].

- **System Integration**: The integration of systems using a chatbot can be used to implement company-wide policies, retrieve current real-time data from back-end systems (CRM systems, Data Repositories).

While these efficiencies are clear and present in existing literature, there are many areas where chatbots need improvement to execute higher level interpersonal skills; specifically those involving empathy and high complexity problems related to consumer complaints [40], [52]. In addition to this, 48% of companies identified "Governance & Data Readiness" as their top obstacle [53].

## V. EVALUATION METRICS

To evaluate Conversational Agents, or Large Language Models (LLMs), a multi-dimensional evaluation method needs to be developed which measures the performance of the algorithms and the user's satisfaction [54] [55]. Since the primary objective of conversational AI is to mimic human interaction while completing tasks, a single measurement will never suffice. As previously mentioned, the majority of research evaluates methods for measuring conversational AI in three different categories: Technical/Linguistic, Functional/Task-Oriented and User-Centered/Experiences [55], [63].

### *A. Technical and Linguistic Metrics*

Automated language assessment metrics including BLEU, ROUGE, and METEOR have typically served as the basis for evaluating the output of Natural Language Generation (NLG). They use n-grams to assess the degree of overlap between a generated response from a machine and a human authored reference response; enabling an extremely scalable and low computational cost approach for determining the level of similarity in two sets of text [56], [63]. Despite this, numerous studies have illustrated the major shortcomings in using overlap based assessments on modern dialogue systems.

### *B. Functional and Task-Oriented Metrics*

In order to evaluate the effectiveness of chatbots at completing tasks related to business operations or service delivery, researchers can utilize what are referred to as functionally derived metrics. Specifically, one of the most important functionally derived metrics is the Task Success Rate; which represents the proportion of all interactions that were fully successful in resolving the user's primary intent [55].

### *C. User-Centric and Experience Metrics*

In addition to providing more accurate measures of performance compared to automated metrics alone, many researchers have also identified the potential to use user-centered metrics to evaluate qualitative dimensions such as fluency, empathy, coherence, and engagingness in conversational systems [58].

#### D. Hybrid Evaluation Frameworks

To help address some of these challenges by developing an evaluation framework that balances the scalability of automated metrics with the accuracy of human judgments, current research emphasizes the need for hybrid evaluation methods [55], [63].

TABLE I  
SUMMARY OF COMMON CHATBOT EVALUATION METRICS

| Metric              | Category     | What it assesses                      | Pros                                                                    | Cons                                                                 |
|---------------------|--------------|---------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| BLEU / ROUGE        | Technical    | N-gram overlap with reference texts   | Highly scalable, objective, and easily computable.                      | Ignores semantics; correlates poorly with human judgment [57], [60]. |
| Hallucination Rate  | Technical    | Factual accuracy and grounding        | Vital for ensuring safety and reliability in LLMs [62].                 | Computationally expensive to verify dynamically.                     |
| Task Success Rate   | Functional   | Completion of the user's primary goal | Directly correlates with operational efficiency and utility [55], [63]. | Difficult to quantify in open-domain conversations.                  |
| Turns to Completion | Functional   | Efficiency of dialogue navigation     | Identifies bottlenecks in conversational flows [59].                    | Shorter interactions aren't always indicative of better quality.     |
| SSA / Engagingness  | User-Centric | Sensibleness and context specificity  | Captures deep conversational quality and human-likeness [61].           | Requires expensive human annotators or advanced LLM judges.          |
| LLM-as-a-Judge      | Hybrid       | Qualitative scoring via advanced AI   | Highly scalable while mimicking human qualitative assessment [58].      | Can suffer from the evaluating model's inherent biases.              |

In conclusion, automated metrics alone are insufficient for evaluating conversational AI. A robust evaluation framework must combine quantitative efficiency with qualitative user experience measures [54], [58], [63].

#### VI. FUTURE RESEARCH DIRECTIONS

Current conversational architectures have advanced to a high level of maturity and while these systems are currently successful in small scale proof-of-concept test environments they must be extended to support enterprise-scale customer service

deployments that will require future engineering efforts in several key areas including system reliability latency reduction and structural data governance for real-time interaction with live consumers.

Architectural scaling is also a critical area of focus for future work as modern RAG-based systems experience practical challenges when it comes to large scale deployment due to massive infrastructure costs and engineering complexity. As such, distributed event stream processing platforms like Apache Kafka which use a publish/subscribe model to handle massive amounts of data in real time will need to become even more commonplace to handle simultaneous customer inquiries during peak hours (e.g., retail holiday spikes or telecom service outages). Because llm inference times can vary from seconds to seconds to seconds to seconds, kafka's asynchronous processing capabilities will provide substantial benefits. Therefore, future research should examine how optimization of scaleable topic partitioning and real time stream processing can ensure continuous reliability and fault tolerance under heavy traffic or backend model outage conditions and prevent any requests made by customers from being dropped.

Operational latency creates another serious bottleneck for live support. The conventional RAG processing pipeline is by design sequential. As such, when processing a user's query, the process involves embedding the user query into a vector database; searching for relevant vectors within said database; and subsequently generating a response to the user. Each stage within this sequence has the ability to create its own latency bottle neck. Due to this sequential nature of processing, it is possible to see the overall latency exceed the expected time budget. For example, real-time or voice enabled AI agents intended to replace existing IVR systems require sub-200 ms response times in order to provide a natural experience for callers. Thus, future research needs to explore various deep optimization techniques including semantic caching for FAQs, and predictive pre-fetching to reduce the operational latency experienced during live chat sessions [29]. In addition to optimizing the processing of user input and responses, there is a need for researchers to begin exploring "vector-less" RAG based approaches. These include hierarchical indexing using methods similar to those employed in PageIndex methodologies [30]. Unlike the method of employing dense vector embeddings to represent document structure, these frameworks will enable improved accuracy for longer and more complex service documents such as banking compliance manuals, warranty policies, and insurance contracts. However, the impact of these frameworks upon real-time inference latency while processing user queries during live chat sessions are areas requiring further empirical investigation [30].

In addition to evaluating performance characteristics, today's chatbots must successfully integrate generative probability-based text output with rigidly defined deterministic enterprise workflows (i.e., refund processes, updates to account status). Additionally, generative RAG models have introduced new surface areas through which malicious intent may be embedded into user input (queries or content), potentially

circumventing enterprise level constraints and providing unauthorized access to business sensitive information such as PII, payment data etc. [64]. Therefore, future architectures must include multiple layers of security to prevent malicious intent from being generated via prompts or other means and must utilize injectors specifically designed to detect attempts to inject malicious intent into generated output in highly reliable environments such as digital banking or health care triage. Finally, investigating the deep integration of structural knowledge graphs with RAG frameworks presents an important path for future research that could help reduce "ai hallucinations" and enforce factual accuracy for complex problem solving in customer service applications [65].

## VII. CONCLUSION

A thorough review of the development of AI-based chatbot technology in customer service has been provided through this survey. The development of the field has moved from a rigid, rules based system with simple search functionality to very flexible and dynamic systems using LLMs and RAG. Through examples of how this technology has been applied in several different areas such as banking, retail and healthcare; it is clear that there has been a significant improvement in response time, accuracy and customer satisfaction. However, the migration from experimental conversational AI to production-ready customer service platforms is primarily an engineering challenge rather than a linguistics challenge. Although the generative model may be able to understand the complexity of what the user is trying to communicate; implementing it into a production environment, which may consist of thousands of users, requires optimizing the underlying infrastructure for operational latency, data governance, and scalability. For example, when providing a deterministic workflow such as processing financial transactions securely or verifying warranty policies accurately, addressing the performance bottleneck associated with all three factors listed above is essential. Ultimately, the success of integrating AI-based chatbots into customer service is dependent on developing hybrid architectures that bridge the gap between the probability-based nature of generated text and the strict reliability requirements of the enterprise. Using continuous refinements of hybrid architectures, continuing to develop evaluation metrics that are beyond traditional n-gram models, and researching new approaches for low-latency retrieval methods will provide companies with the ability to create a robust and reliable support platform. When implemented correctly, AI-based chatbots should be able to efficiently process routine, data intensive questions allowing human customer service personnel to focus their efforts on providing empathy and resolving the most complex issues, which is still key to delivering superior levels of customer care.

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